SAMPLE QUESTIONS 2B2 (MULTIPLE CHOICE)

(NOTE: These questions are intended as representations of the style of questions that may appear on examinations, they are not intended as study material and, as such, may not be in line with any current examination syllabus)

Currently all the 2B2 SOPEEC examinations are delivered in a multiple-choice format. Examination candidates have 3.0 hours to answer 100 multiple choice questions. This works out to be 1.8 minutes per question.

To use your time wisely it is recommended that you read through the questions and answer only the questions for which the answer comes to you quickly. Leave other questions and calculation questions for later.

High levels of iron in the boiler water and in the blowdown are a sign of

A. excessive blowdown.
B. a tube leak in the condenser.
C. corrosion in the boiler feedwater system.
D. continuous blowdown piping being plugging off.

Low pump suction pressure in a fuel-oil handling system would most likely indicate that the

A. pump PSV is passing.
B. suction strainers need cleaning.
C. discharge strainers are plugging off.
D. line is plugged downstream of the pump.

What are the mechanisms employed to reduce the water's temperature in an open-circulating cooling water system?

A. Induced or balanced draft fans.
B. Evaporative cooling and contact with air.
C. Cooling towers and absorption refrigeration systems.
D. Cooling towers and compression refrigeration systems.

What is a key consideration of the use and placement of flow measuring devices?

A. Downstream taps should always be mounted on the top of the pipe or duct to avoid plugging.
B. Upstream taps should always be mounted on the top of the pipe or duct to avoid plugging.
C. Most flow-measuring elements require a minimum length of straight run piping both upstream and downstream of the measurement point.
D. The taps need to be kept as far apart as possible to ensure turbulence caused by upstream port does not affect downstream reading.

A _____ valve cage is designed to have a sharp increase in flow with little plug travel form opening.

A. low-noise trim
B. quick opening
C. linear opening
D. equal percentage

When heat is given off as a by-product of the chemical reactions, the combustion process is said to be

A. adiabatic
B. exothermic
C. endothermal
D. endothermic

Which of the following is an electrical device that uses a microprocessor to monitor data inputs and outputs and controls the switching of the outputs according to software instructions contained in its user program?

A. Fieldbus network controller
B. Building management system
C. Input output module controller
D. Programmable logic controller

What is the principle component of natural gas?

A. Butane
B. Ethane
C. Propane
D. Methane

A venture-type design for air admittance would most commonly be found on a _____ grate system.

A. pinhole
B. vibrating
C. travelling
D. stationary

When firing liquid fuel, the metering orifice is often combined with a section that allows a/an _____ to be mixed with the fuel in order to break it to droplets.

A. additive
B. oxidizing agent
C. atomizing medium
D. liquefied natural gas

In an electrostatic precipitator, what is the ability of a flyash particle to accept an electric charge called?

A. Resistivity
B. Magnetism
C. Electrolysis
D. Back corona

Which industrial pollutant, when present in water bodies as a contaminant, encourages excessive growth of water plants, thus alters the natural habitat?

A. Iron
B. Phosphorous
C. Hydrocarbons
D. Sulphur dioxide